# Yucheng Yin (阴煜程)

yyin@caltech.edu cuberyyc808.github.io

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## Education

**California Institute of Technology** **Pasadena, CA, USA**
Ph.D. in Physics 09/2026 – Present

**School of Physics, Peking University** **Beijing, China**
Bachelor's Degree of Science 09/2021 – 07/2025

- Major in Physics (Astrophysics)
- Research interests: gravitational wave astrophysics, black hole perturbation theory.
- Thesis: Gravitational radiation of binary extreme-mass-ratio inspiral systems (Click **here**)

## Research Experience

**California Institute of Technology** **Pasadena, CA, USA**
Graduate researcher 2026 – Present

**Advisor**: Prof. Yanbei Chen

**Niels Bohr Institute, University of Copenhagen** **Copenhagen, Denmark**
Visiting student 10/2025 – 03/2026

**Advisor**: Dr. Rico K. L. Lo

**Department of Physics, National University of Singapore** **Singapore**
Summer intern 06/2024 – 08/2024

**Advisor**: Prof. Alvin J. K. Chua

**Kavli Institute for Astronomy and Astrophysics, Peking University** **Beijing, China**
Undergraduate researcher 02/2023 – 06/2025

**Advisor**: Prof. Xian Chen

## Skills

- Programming Languages: Mathematica $\sim$ Julia $\sim$ Python > C
- BHPT: RWZ formalism, Teukolsky formalism, Sasaki-Nakamura formalism, QNMs theory, Self-force, ...
- Others: Numerical methods to solve ODE and PDE

## Professional Membership

LISA (Laser Interferometer Space Antenna) consortium, **Core** member 2025 – Present

Waveform working group - working on EMRI 0PA flux dataset generation and validation

## Tools and Codes

- `KerrGeodesics.jl`: Solves timelike Kerr geodesics.
- `AdaptiveLevin.jl`: Evaluates one- and two-dimensional highly oscillatory integrals.
- `GeneralizedSasakiNakamura.jl`: Solves the radial Teukolsky equation.
- `SpinWeightedSpheroidalHarmonics.jl`: Solves the angular Teukolsky equation.

## Talks

**Contributed talks**

BIMSA Workshop on Gravitational Wave Astronomy, Beijing 04/2025

**Invited talks**

School of Physics and Astronomy, Sun Yat-sen University 11/2025

## Honors

Outstanding Undergraduate Thesis from Peking University 2025

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## References

**Vitor Cardoso**
*Professor, University of Copenhagen & Universidade de Lisboa*

**Alvin J. K. Chua**
*Assistant Professor, National University of Singapore*

**Xian Chen**
*Associate Professor, Peking University*